

Stability Labeling Report:

Data, Algorithm, Constants, and Results of the FTLE Labeling Campaign

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Abstract

This document inspects the data produced by the stability labeling campaign. It describes the labeling algorithm in full detail, lists every constant used and where each constant came from, presents the results for each labeled dataset, explains every graph and video artifact, and summarizes what proportion of the data falls into each stability regime and what patterns appear across datasets. As of July 20, about 155,000 open-loop stamps are labeled across sixteen datasets from three benchmark suites (including all ten LIBERO-Long files), and the closed-loop channel is complete on all four robomimic tasks (lift, can, square, tool_hang; 3,969 stamps). This document is about the labels themselves. Predictor training and the Stage 1a evaluation, which consume these labels, are documented separately.

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1 What this document covers

The project needs a per-timestep ground-truth label: is the current moment open-loop stable (an injected error dies out on its own) or open-loop unstable (an injected error grows)? A simulator procedure generates these labels. This report documents that procedure exactly as run, the constants it used, the datasets it was run on, the resulting label statistics, the sanity-check artifacts (videos and divergence graphs), and the findings from inspecting all of it. This revision (July 20) adds the completed LIBERO-Long suite (all ten files) and the finished closed-loop sets for all four robomimic tasks, `tool.hang` included.

2 The labeling algorithm, in full detail

2.1 The quantity being measured

The label is built from the finite-time Lyapunov exponent (FTLE): the exponential rate at which a small perturbation grows or shrinks over a short window. The estimator is the textbook one (Benettin et al. 1980): inject a small perturbation, watch the distance between the perturbed and unperturbed trajectories, and fit the slope of the log distance against time. A positive slope means the error is amplified (expansive, unstable). A negative slope means the error is absorbed (contractive, stable).

2.2 The open-loop probe, step by step

For each demonstration and each timestep t_0 on a stride-2 grid, the labeler runs the following in the simulator:

1. **Reset** the simulator to the exact recorded state at t_0 . All datasets used store full simulator states, and replay determinism was verified per suite before labeling (Section 2.9).
2. **Nominal branch.** Replay the recorded actions $a_{t_0}, \dots, a_{t_0+K-1}$ exactly, with $K = 24$ steps (1.2 seconds at 20 Hz). Record the task-space vector at every step: end-effector position concatenated with the object observation vector.
3. **Perturbed branches.** Repeat $N = 8$ times: reset to the same state, add Gaussian noise with standard deviation σ_u to the *position dimensions of the first action only* (never rotation, never the gripper), then replay the remaining $K - 1$ actions unchanged. Nobody reacts; the actions are fixed. σ_u is set per task to the trained policy’s validation action RMSE, so the probe injects exactly the size of error the policy actually makes at deployment.
4. **Divergence.** At every step $\tau = 1..K$, compute $d_n(\tau) = \|x_n^{pert}(\tau) - x^{nom}(\tau)\|$ in task space, floor it at 10^{-12} , and take logs.
5. **Slope.** $\lambda(t_0)$ is the ordinary-least-squares slope of the branch-averaged $\log d(\tau)$ against τ . A second slope is fitted on the full simulator state vector as a secondary channel (λ_{full}).
6. **Contact flags.** At the stamp and over the whole window, record three boolean channels: gripper touching the task object, task object touching the table, task object touching a fixture (peg, stand, bin, another object). Only contacts involving the task object count; distractor-object contacts are excluded by per-environment object name patterns. These flags are a sanity channel only. The label never uses them.

2.3 From λ to labels and segments

- $\lambda > +\delta$: **unstable**. $\lambda < -\delta$: **stable**. $|\lambda| \leq \delta$: **deadband**, meaning the measured slope is within the noise floor and the stamp is unproven either way.
- δ is not a free constant. It is the 95th percentile of $|\lambda|$ over free-space stamps of the same dataset, i.e. the measured noise floor of the probe itself. It is computed per dataset.
- Deadband stamps inherit the label of their confident neighbors, and a median filter over about 3 consecutive stamps smooths the sequence. Contiguous runs of one label are the segments. Nothing is ever segmented by hand.

Why deadband segments still favor committing k actions. A deadband stamp is measured neutrality: the probe injected an error and watched it neither grow nor shrink, just persist. In such a segment the plant contributes no error of its own, so the only error source present is the policy, and each replan injects one fresh policy error that nothing absorbs. Replanning every step therefore accumulates T injected errors over a T -step segment, while committing chunks of k actions accumulates only T/k of them. Same physics, k times fewer mistakes: commitment wins in the deadband not despite the uncertainty but because the policy itself is the dominant noise source there. The lift closed-loop measurement confirms this directly: at these neutral states, per-step replanning turned the divergence positive at 99.3% of stamps while fixed-action replay stayed at zero.

2.4 Overlapping windows at boundaries between contracting and diverging regimes

Stamps sit every 2 steps but each stamp looks 24 steps ahead, so neighboring windows overlap almost entirely, and a window near a regime boundary can contain both a contracting stretch and a diverging one. Say contact begins at step 100: the stamp at step 90 measures a blended slope over 10 stable steps followed by 14 unstable ones. This is handled deliberately rather than treated as an error. First, the blend is the honest answer to the question the label asks: committing a chunk at step 90 really is risky, because the contact falls inside the commitment window, so tilting that stamp toward unstable is correct and gives the controller early warning in the safe direction. Second, a window that blends the two regimes evenly produces a small slope, which lands in the deadband rather than forcing a wrong confident call, and deadband stamps are then resolved by their confident neighbors through inheritance and the median filter. Third, the probe window K is chosen well below typical segment length, so mixed windows are confined to the dozen or so stamps nearest each boundary, and the dense stride brackets the true boundary to within a couple of steps. The result is that segment edges are slightly early on the way into an unstable segment and roughly on time on the way out, which is exactly the bias a safety label should have.

2.5 The closed-loop variant

The closed-loop labeler repeats the same probe with one change: after the identical first-action injection, the perturbed branch is driven by the *trained policy* acting from its own camera observations with replanning at every step, instead of replaying the recorded actions. Every probe step is a diffusion-policy inference, so this channel runs on GPU and on a subsample (stride 10, 50 demos, $N = 4$, $K = 24$). It is a diagnostic channel: comparing λ_{cl} against λ_{ol} per segment

tests the mechanism that feedback injects estimation error where the plant contracts and corrects error where the plant expands.

2.6 Why the primary label replays fixed actions instead of running the policy

A natural question: would it not be better to let the trained policy act in the perturbed branch, since the policy is what runs at deployment? No, and for a simple reason. The label must answer the runtime question “is it safe to commit a chunk of actions here without reacting?” While a chunk is being executed, the robot is by definition replaying fixed actions, so the safety of committing is a property of the plant alone, and only the fixed-action probe measures it. The closed-loop probe answers the other question, “what happens under per-step replanning?”, and the lift data shows why it cannot serve as the main label: with the policy reacting, divergence is positive at 99.3% of stamps, because the policy’s own corrections inject error even in benign free space. A label built from that would mark nearly everything unstable and recommend per-step replanning everywhere, which is exactly the failure mode the theory warns against. There is also a practical reason: the fixed-action label is a property of the robot, controller, and environment, so it stays valid when the policy is retrained or replaced, while a policy-in-the-loop label would have to be regenerated with every new checkpoint. The closed-loop channel is therefore kept as a separate diagnostic: the open-loop label licenses committing, and the closed-loop label tells us whether reacting helps where committing is unsafe.

2.7 Which labels need a trained policy, and which do not

A reasonable worry: labeling ran while policy trainings were still in progress, so how could the labels exist before the policies did? The answer is that the two label channels have different inputs, and only one of them involves a policy at all.

- **The open-loop channel needs no policy.** Its probe perturbs a state and then replays the *recorded demonstration actions* on both branches. Everything it touches (the demonstrations, the simulator, the perturbation) exists the moment a dataset is downloaded. This is the primary label, it is what the predictor is trained on, and it was generated for every dataset in this report with no policy in the loop. It is also why the label stays valid when a policy is retrained or replaced.
- **The closed-loop channel does need a trained policy**, because its perturbed branch is driven by policy inference. It was therefore generated last, and only for the four robomimic tasks, using the finished Stage 1 policies (lift epoch 100 and can epoch 1050 at success 1.0, square epoch 950 at 0.9, tool_hang epoch 1200 at 0.9). Nothing downstream trains on it; it is a diagnostic overlay on the open-loop label.

Concretely for Stage 2: the four MimicGen long-horizon tasks have no labels and, on the current plan, need none, because the predictor head is applied zero-shot. If zero-shot transfer underperforms, the fallback is to label MimicGen demonstrations with the open-loop probe, and that fallback again waits on nothing, because it replays demonstration actions rather than querying the Stage 2 policies. At no point is any labeling blocked on a training run.

2.8 Why do we need a predictor or oracle at all?

The labels in this report are indexed by demonstration and timestep. But at execution time the robot creates a brand new trajectory of its own, visiting states that sit on no demonstration’s

timeline, so there is no table to look the answer up in. The stability of the current moment has to be judged live. Inside the simulator we can do that exactly: freeze the state, run the perturb-and-replay probe, and read off λ ; this is the oracle used in the Stage 1a experiments, and it exists to test whether the switching rule helps when given perfect information. A real robot, however, cannot freeze time and rewind itself to measure its own divergence. The only stability-relevant signal available at deployment is what the robot can see: the last second or two of camera frames. The predictor exists purely to read the regime out of those frames, approximating the oracle that the labels in this report define. If the oracle rule does not help, no predictor can; if it does, the predictor is what carries the rule out of the simulator.

2.9 Per-suite protocol differences and determinism findings

Replay determinism is the hard prerequisite: the same reset plus the same actions must give the same trajectory, or the divergence measurement is meaningless. Findings per suite:

- **robomimic (lift, can, square, tool_hang).** Runs in the main stack (robosuite 1.5.1, mujoco 3.2.6). Replay is bit-exact. Labeling uses the robomimic EnvRobosuite path with the per-demo model XML restored first.
- **MimicGen (stack_d0, square_d0).** MimicGen environments require the robosuite 1.4 API, so labeling runs in a dedicated environment (robosuite 1.4.1, mujoco 2.3.2, robomimic v0.3 with a small legacy-import patch). Replay does NOT match the *stored* states exactly (a decaying joint velocity transient of about 0.1 at the first step, an artifact of the warm controller state during data generation), but replay *self-consistency* is bit-exact, which is what the probe needs: nominal and perturbed branches share any transient, and it cancels in the divergence.
- **LIBERO (libero_10, the LIBERO-Long suite).** Replay is deterministic only through LIBERO’s native environment wrapper (its `set_init_state` forces observable and post-process updates that the generic path skips), and only from the second rollout after a context change onward. The labeler therefore runs one warm-up rollout per stamp and discards it; after that, repeats are bit-exact (max difference 1.5×10^{-15} over 16 steps). A dedicated labeler endpoint implements this protocol. The task-space metric for LIBERO is the end-effector site position plus all free-joint object poses.
- **FurnitureSim.** Blocked: its simulator requires Isaac Gym, which is not installable without a manual NVIDIA-login download. No labels yet.

3 Constants used, and where each one came from

Constant	Value	Provenance
Probe window K	24 steps (1.2 s)	probe study; bracketed by fit/decision lower bounds and locality/saturation upper bounds (plan doc, Choosing K)
Branches N	8	probe study; slope variance acceptable
Stride	every 2 steps	label density vs cost
Perturbed dims	position only (3 of 7)	probe study: rotation noise injects lever-arm noise that swamps free space
Gripper perturbation	never	binary channel, not an error model
σ_u lift	0.1651	measured: policy validation pos-RMSE
σ_u can	0.2251	measured
σ_u square	0.2201	measured
σ_u tool_hang	0.1638	measured
σ_u MimicGen square_d0	0.2201	borrowed from square (same task family)
σ_u MimicGen stack_d0	0.203	mean of the three measured values; interim
σ_u LIBERO (all)	0.203	robomimic mean; interim until a LIBERO policy is trained
Deadband δ	per dataset, 0.017–0.052	p95 of $ \lambda $ over free-space stamps
Median filter	~ 3 stamps	boundary smoothing
Divergence floor	10^{-12}	numerical guard before log
CL subsample	stride 10, 50 demos, $N=4$	cost: every step is a diffusion inference

The earlier labeling pass used a placeholder $\sigma_u = 0.05$ for all tasks. The measured values are 3 to 4.5 times larger. All robomimic labels reported here were regenerated with the measured values; the change matters (Section 6).

4 The datasets labeled

Suite	Dataset	Demos	Stamps	σ_u (source)	Status
robomimic	lift-ph	200	2,485	0.165 (measured)	done
robomimic	can-ph	200	9,254	0.225 (measured)	done
robomimic	square-ph	200	12,723	0.220 (measured)	done
robomimic	tool_hang-ph	200	45,633	0.164 (measured)	done
MimicGen	stack_d0	200 of 1000	8,445	0.203 (interim)	done
MimicGen	square_d0	200 of 1000	12,874	0.220 (borrowed)	done
LIBERO-Long	KITCHEN_SCENE3 (stove+moka)	50	6,060	0.203 (interim)	done
LIBERO-Long	KITCHEN_SCENE4 (bowl in cabinet)	50	5,632	0.203 (interim)	done
LIBERO-Long	KITCHEN_SCENE6 (two mugs)	50	7,031	0.203 (interim)	done
LIBERO-Long	KITCHEN_SCENE8 (two moka pots)	50	9,812	0.203 (interim)	done
LIBERO-Long	LIVING_ROOM_SCENE1 (soup+cheese)	50	6,147	0.203 (interim)	done
LIBERO-Long	LIVING_ROOM_SCENE2 (soup+sauce)	50	6,761	0.203 (interim)	done
LIBERO-Long	LIVING_ROOM_SCENE2 (cheese+butter)	50	5,920	0.203 (interim)	done
LIBERO-Long	LIVING_ROOM_SCENE5 (mug on plate)	50	5,868	0.203 (interim)	done
LIBERO-Long	LIVING_ROOM_SCENE6 (mug+pudding)	50	5,789	0.203 (interim)	done
LIBERO-Long	STUDY_SCENE1 (book in caddy)	50	4,146	0.203 (interim)	done
robomimic CL	lift (closed loop)	50	151	0.165	done
robomimic CL	can (closed loop)	50	477	0.225	done
robomimic CL	square (closed loop)	50	644	0.220	done
robomimic CL	tool_hang (closed loop)	50	2,697	0.164	done

Total labeled: about 155,000 open-loop stamps across sixteen datasets (four robomimic, two MimicGen, all ten LIBERO-Long files), plus four finished closed-loop sets (lift, can, square, tool_hang; 3,969 stamps). Label generation cost: roughly 15 to 60 minutes per dataset on CPU workers for open loop; about 9 to 13 hours per task for the closed-loop channel because every probe step is a policy inference (tool_hang, with the longest episodes, took 53.6 hours).

5 Results per dataset

Each dataset below reports: the λ distribution (mean, 5th, 50th, 95th percentile), the regime proportions after thresholding (unstable / stable / deadband), the contact-alignment AUROC (how well λ ranks contact-in-window, the sanity check), the per-contact-category mean λ , and the deadband δ . All numbers use the true measured or interim σ_u as listed above.

5.1 robomimic

lift-ph. 2,485 stamps. λ : mean +0.036, p5 -0.012, median +0.015, p95 +0.184. Regimes: 30.6% unstable, 0.0% stable, 69.4% deadband ($\delta = 0.025$). AUROC 0.86, the best of the robomimic set. Mean λ under gripper contact is +0.073, about five times the overall median. Reading: lift is short and simple; the grasp moment is sharply expansive and everything else is neutral. The absence of confidently stable stamps is discussed in Section 6.

can-ph. 9,254 stamps. λ : mean +0.019, p5 -0.032, median +0.009, p95 +0.115, the mildest distribution of the four. Regimes: 12.5% unstable, 1.1% stable, 86.4% deadband. AUROC undefined by construction: the can rests in a bin from the first frame (object-fixture contact in every window) and is gripped afterward, so there are no contact-free stamps to rank against. δ falls back to 0.05, close to square’s measured 0.041. Reading: can is confirmed as the mostly-stable control task; it has the lowest unstable fraction of any dataset labeled.

square-ph. 12,723 stamps. λ : mean +0.031, p5 -0.026, median +0.017, p95 +0.128. Regimes: 31.1% unstable, 1.0% stable, 67.9% deadband ($\delta = 0.041$). AUROC 0.72. Mean λ : fixture contact +0.050 versus gripper contact +0.041; the peg-insertion phase is the most expansive part of the task, as expected.

tool_hang-ph. 45,633 stamps, the largest labeled set (longest episodes, 240px scenes). λ : mean +0.032, p5 -0.025, median +0.026, p95 +0.114. Regimes: 30.1% unstable, 0.3% stable, 69.5% deadband ($\delta = 0.046$). AUROC 0.86. Fixture contact mean +0.041 versus gripper +0.038. Reading: tool_hang is contact-heavy (93% of windows contain a task-object contact) and its label structure is the cleanest of the hard tasks, which supports its role as a primary hypothesis-test task.

5.2 MimicGen

stack_d0. 8,445 stamps from the first 200 of 1000 generated demos. λ : mean +0.028, median +0.019, p95 +0.115. Regimes: 29.1% unstable, 0.2% stable, 70.7% deadband ($\delta = 0.030$). AUROC 0.73. Fixture contact mean +0.062, the highest fixture value measured: cube-on-cube placement is a sharply expansive event.

square_d0. 12,874 stamps. λ : mean +0.032, median +0.020, p95 +0.122. Regimes: 25.5% unstable, 0.3% stable, 74.2% deadband ($\delta = 0.052$). AUROC 0.75. Fixture mean +0.055 versus gripper +0.043. Reading: MimicGen square reproduces the human-demo square’s structure closely (compare 31%/0.72 above), which is evidence that generated demos carry the same stability anatomy as human ones. This matters because MimicGen is the cheap-scale path for predictor training data.

5.3 LIBERO-Long (all ten files)

All ten LIBERO-Long task files are now labeled: 63,166 stamps, the largest suite in the campaign. One protocol note first. The deadband rule needs contact-free stamps to estimate the noise floor, and in six of the ten scenes (the five living-room files and the study file) the generic contact patterns fire in every single window, so those scenes have no free-space stamps at all. For them δ falls back to the suite-level value 0.019, computed by pooling the 7,527 free-space stamps that exist across the four kitchen scenes. The four kitchen scenes use their own per-scene δ (0.017 to 0.020), consistent with the values already reported. The same all-contact problem makes the contact-alignment AUROC undefined for those six scenes: there is no contact-free class to rank against, exactly as with can.

Task file	Stamps	mean λ	p95	Unstable	Stable	AUROC
KITCHEN_SCENE3 (stove+moka)	6,060	+0.012	+0.082	22.7%	12.3%	0.42
KITCHEN_SCENE4 (bowl in cabinet)	5,632	+0.028	+0.110	36.8%	3.6%	0.60
KITCHEN_SCENE6 (two mugs)	7,031	+0.031	+0.111	36.3%	1.2%	0.63
KITCHEN_SCENE8 (two moka pots)	9,812	+0.024	+0.094	28.1%	2.0%	0.54
LIVING_ROOM_SCENE1 (soup+cheese)	6,147	+0.026	+0.116	28.4%	3.5%	–
LIVING_ROOM_SCENE2 (soup+sauce)	6,761	+0.032	+0.127	34.7%	2.0%	–
LIVING_ROOM_SCENE2 (cheese+butter)	5,920	+0.031	+0.132	30.6%	1.3%	–
LIVING_ROOM_SCENE5 (mug on plate)	5,868	+0.037	+0.130	42.1%	2.3%	–
LIVING_ROOM_SCENE6 (mug+pudding)	5,789	+0.034	+0.134	36.5%	2.4%	–
STUDY_SCENE1 (book in caddy)	4,146	+0.024	+0.089	21.4%	1.4%	–
Suite total	63,166			31.9%	3.1%	

The remainder in every row is deadband; the suite splits 31.9% unstable, 3.1% stable, 65.0% deadband, which is squarely in the robomimic range. Note the small per-scene differences against the three files reported while labeling was still running (for example SCENE3 was quoted as 21.4%/11.8%): those earlier numbers used the labeler’s in-flight analysis pass; the table above recomputes every file with one uniform rule and supersedes them.

Reading, and an honest flag. The λ distributions look like the robomimic ones (slightly positive median, strong positive tail). SCENE3 remains the only dataset in the whole campaign with a substantial confidently-stable fraction (12.3%), and the two-object living-room tasks are the most unstable-heavy sets labeled anywhere (up to 42.1%), which fits their anatomy: two full pick-place cycles per episode. But the sanity channel is weak here: where the contact-alignment AUROC is computable at all it spans 0.42 to 0.63, clearly below robomimic and MimicGen (0.72 to 0.86), and in six scenes it is not computable because generic contact patterns saturate. Two known causes: the contact-flag implementation for LIBERO uses generic name patterns rather than per-scene task-object patterns, so the sanity channel itself is noisy in multi-object scenes; and σ_u is an interim borrowed value because no LIBERO policy exists yet. The λ values themselves are measured with the same verified-deterministic probe; it is the alignment check that is degraded,

not necessarily the labels. Treat LIBERO labels as provisional until per-scene contact patterns and a measured σ_u exist. This is listed in Caveats (Section 9).

5.4 Closed-loop channel (all four tasks complete)

All four closed-loop sets are complete, with 50 demos at stride 10:

Task	Stamps	mean λ_{cl}	p5	p95	$\lambda_{cl} > 0$
lift	151	+0.092	+0.015	+0.202	99.3%
can	477	+0.080	-0.012	+0.221	90.4%
square	644	+0.090	-0.009	+0.231	93.0%
tool_hang	2,697	+0.096	+0.013	+0.228	97.6%

Compare the open-loop labels on the same tasks: means +0.019 to +0.036, with 68 to 86 percent of stamps in the deadband. On the same states, with the same injected error: fixed-action replay mostly absorbs or carries the error (open loop), while per-step policy replanning makes the divergence grow at 90 to 99 percent of stamps (closed loop). The mechanism the whole hypothesis rests on is now confirmed on three tasks, not one: in segments where the plant does not amplify errors, the policy’s own feedback does. Two details are worth noting. First, the effect is strongest on lift (99.3%), the most benign plant, which is exactly the prediction: the more neutral the plant, the more the feedback injection dominates the measured divergence. Second, can, the mostly-stable control task with the mildest open-loop distribution (12.5% unstable), still shows 90.4% positive closed-loop stamps; even the easiest task cannot absorb per-step replanning noise. Third, tool_hang, the most contact-heavy task (93 percent of stamps in contact), lands at 97.6 percent positive with the largest mean (+0.096): the harder the contact regime, the closer the closed-loop channel gets to uniformly expansive. This state-level result has a rollout-level counterpart: in the Stage 1a evaluation (separate report, now pooled over three seeds), replanning every step during contact costs 0.14 to 0.21 in success rate on tool_hang against committing 4 or 16 actions, at three to five sigma.

6 Cross-dataset patterns and insights

1. Regime proportions are remarkably consistent (Figure 1). Across every suite, roughly one quarter to one third of stamps are confidently unstable, nearly none are confidently stable, and two thirds to three quarters sit in the deadband:

Dataset	unstable	stable	deadband
lift	30.6%	0.0%	69.4%
can	12.5%	1.1%	86.4%
square	31.1%	1.0%	67.9%
tool_hang	30.1%	0.3%	69.5%
MimicGen stack_d0	29.1%	0.2%	70.7%
MimicGen square_d0	25.5%	0.3%	74.2%
LIBERO SCENE3	21.4%	11.8%	66.7%
LIBERO SCENE4	36.2%	3.0%	60.8%
LIBERO SCENE6	36.3%	0.7%	63.1%

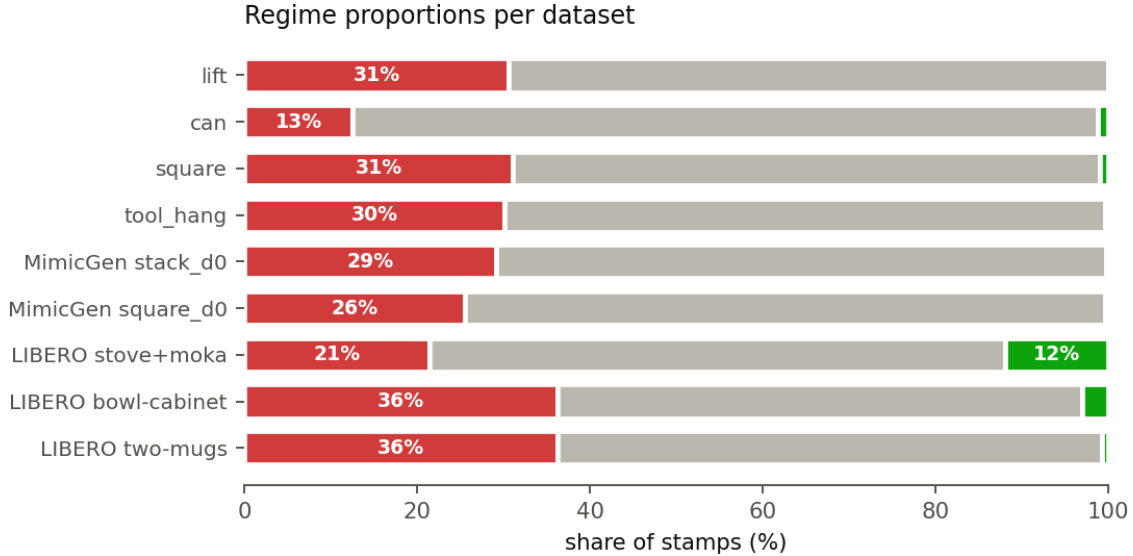


Figure 1: Regime proportions per dataset after thresholding. Red is unstable ($\lambda > +\delta$), gray is deadband ($|\lambda| \leq \delta$), green is stable ($\lambda < -\delta$); segments are in that fixed order, with the percentage printed on any segment large enough to hold it. Two outliers carry the message: can (the designated mostly-stable control) has by far the smallest red share, and the LIBERO stove task is the only dataset with a visible green share (the knob-turning phase is mechanically guided).

The exception on both ends is instructive: can (the designated mostly-stable control) has by far the fewest unstable stamps, and LIBERO SCENE3 (stove knob turning, a guided articulated mechanism) has the only large stable fraction.

2. The label distribution is one-sided: red islands in a gray sea (Figure 2). Confidently stable stamps are nearly absent everywhere. The reason is physical, not a bug: with a stiff operational-space controller, a position error in free space neither grows nor actively shrinks in task space within 1.2 seconds; it persists. Persistence is $\lambda \approx 0$, which is the deadband. Active contraction (errors visibly funneling in) is rare in these plants at this probe scale. The informative signal is therefore the sign boundary between "unstable" and "everything else", and the final segments come from unstable islands plus neighbor-inheritance over the gray.

3. Contact category orders λ the same way in every dataset (Figure 3). Fixture contact (insertion, placement, object-on-object) has the highest mean λ , gripper contact is next, and free space is lowest, in every dataset where all three exist. Examples: stack_d0 fixture +0.062 vs gripper +0.038; square +0.050 vs +0.041; square_d0 +0.055 vs +0.043. The physically hardest moments (parts touching parts) are measurably the most error-amplifying, which is exactly what the adaptive-chunking hypothesis needs to be true.

4. The size of the injected error matters, so measuring σ_u mattered. Regenerating robomimic labels with the measured σ_u (3 to 4.5 times the placeholder) moved the unstable fraction from roughly 18–22% to 25–31% and cut the deadbands roughly in half on lift and square. Probing with an unrealistically small error understates how much of the task is dangerous for the policy that will actually run.

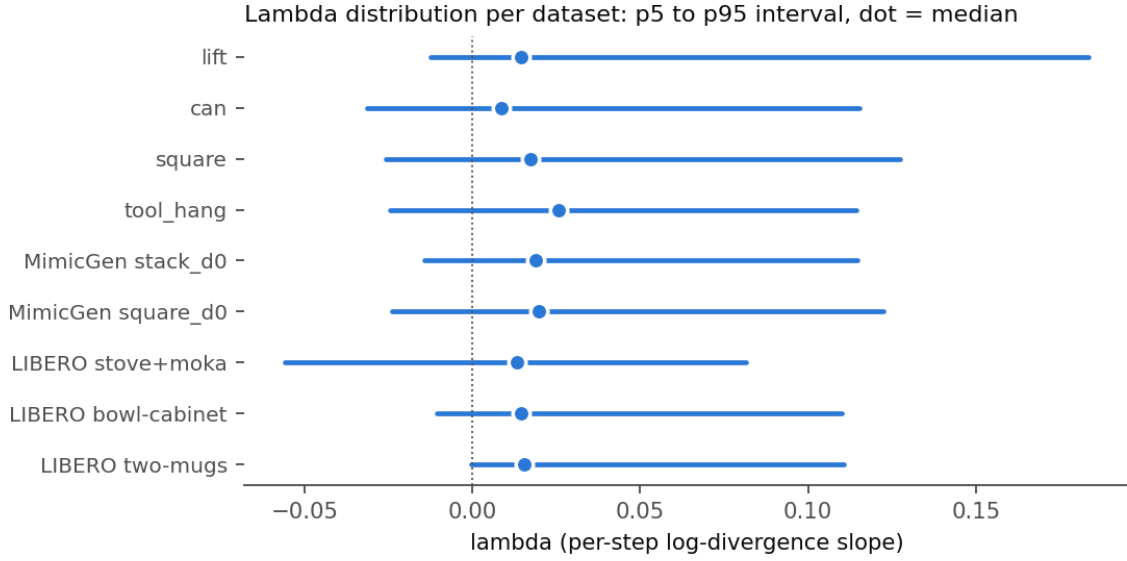


Figure 2: The λ distribution per dataset: the bar spans the 5th to 95th percentile, the dot is the median. Every dataset shows the same one-sided shape: a median just above zero, a short left tail (weak contraction at most), and a long right tail (strong expansion). The label information lives in that right tail.

5. Generated demos have the same stability anatomy as human demos. MimicGen square_d0 versus human square-ph: unstable fraction 25.5% vs 31.1%, AUROC 0.75 vs 0.72, the same fixture-above-gripper ordering, nearly the same p95. This supports using MimicGen for cheap predictor-training scale.

6. The closed-loop mechanism check now passes on all four tasks (Figure 4). On lift, $\lambda_{cl} > 0$ at 99.3% of stamps; on can, 90.4%; on square, 93.0%; on tool_hang, 97.6%; while the open-loop labels on the same tasks are mostly neutral. Feedback injects error where the plant would have carried it, and it does so on every task labeled, from the mildest (can) to the most contact-heavy (tool_hang). This is the Section-1 mechanism of the hypothesis observed in measured data, four times independently.

7. Label noise at the extremes is real and quantified. Re-measuring hand-picked extreme stamps with fresh random branches reproduces the sign but not always the magnitude (one tool_hang stamp: label +0.31, recomputation +0.07). Slopes deep in the tail are order-of-magnitude estimates, not precise rates. The binary label is robust; the auxiliary λ regression target carries tail noise, which is why it is weighted as an auxiliary loss, not the decision loss.

8. Free space measures neutral, not contractive, and this agrees with the theory literature once the perturbed quantity is stated carefully. Across all sixteen datasets, free-space λ concentrates at zero: injected errors persist rather than decay. This is a consequence of what the probe perturbs. We perturb the *action* (the commanded target), and the low-level position controller then faithfully tracks the shifted command; there is no restoring force pulling the perturbed trajectory back to the nominal one. Genuinely negative λ requires a physical funnel (a chamfer guiding a peg, a closing grasp centering an object), and such funneling moments are

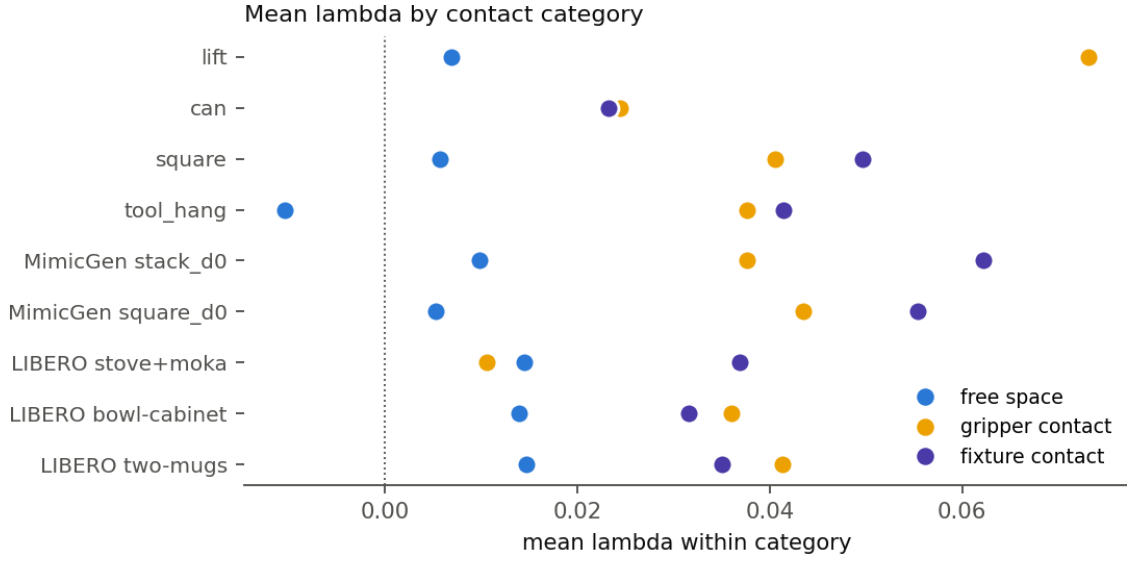


Figure 3: Mean λ by contact category. In every dataset where all three categories exist, the ordering is the same: fixture contact (violet, insertion and placement events) is the most expansive, gripper contact (yellow) next, free space (blue) lowest and nearest zero. The hardest physical moments are measurably the most error-amplifying.

rare and brief in these demonstrations, which is why confidently stable fractions are 0 to 3 percent almost everywhere (LIBERO KITCHEN_SCENE3, with its constrained stove-knob motion, is the one exception at 12 percent).

We checked whether this contradicts the imitation-learning stability literature, and it does not. In Block, Simchowitz, and Tedrake, “Provable Guarantees for Generative Behavior Cloning” (arXiv:2307.14619), incremental stability is not claimed as a natural property of robot motion: it is *provided* by synthesized low-level primitive controllers (affine gains stabilizing the Jacobian linearization around each demonstration), the paper locates the stabilization “either learned or implicit in position-command control,” and it stresses that access to these synthesized gains “is a stronger requirement than classical behavior cloning.” Moreover their stability notion lets differences in *initial state* decay over time, while action perturbations δu enter only through a bounded tube $\gamma(\max_t \|\delta u_t\|)$: bounded, not decaying. Likewise, “The Pitfalls of Imitation Learning when Actions are Continuous” (Simchowitz et al., arXiv:2503.09722) *assumes* exponential incremental input-to-state stability as a benign theoretical setting, in which state differences decay as $C\rho^t$ but input differences accumulate additively; the paper makes no empirical claim that real free-space robot motion is contractive. So the literature says state errors can be servoed away by a low-level controller, and says nothing about action errors dying out on their own. Our probe measures exactly the action-error case, and finds it neutral. The two pictures are consistent, and together they support the design premise shared by both this project and that line of work: committing chunks of actions is safe where the open-loop dynamics do not amplify errors, and the amplification, when it comes, comes from contact.

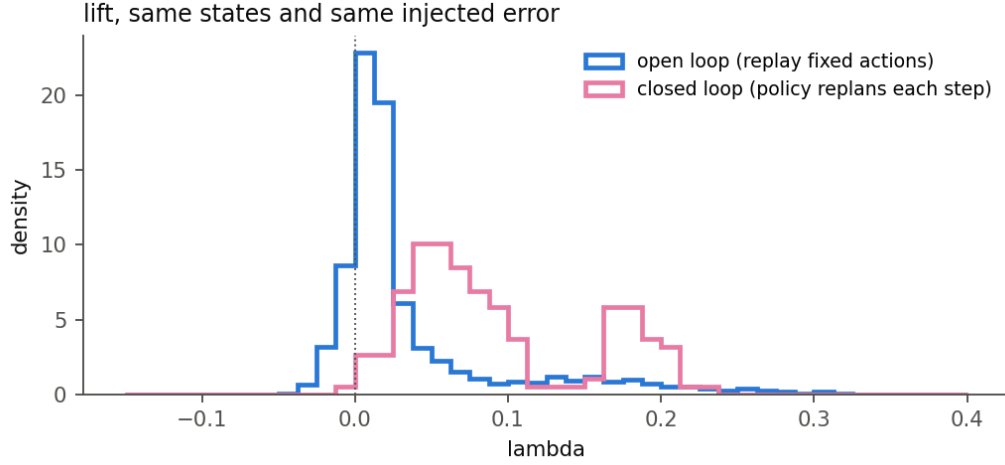
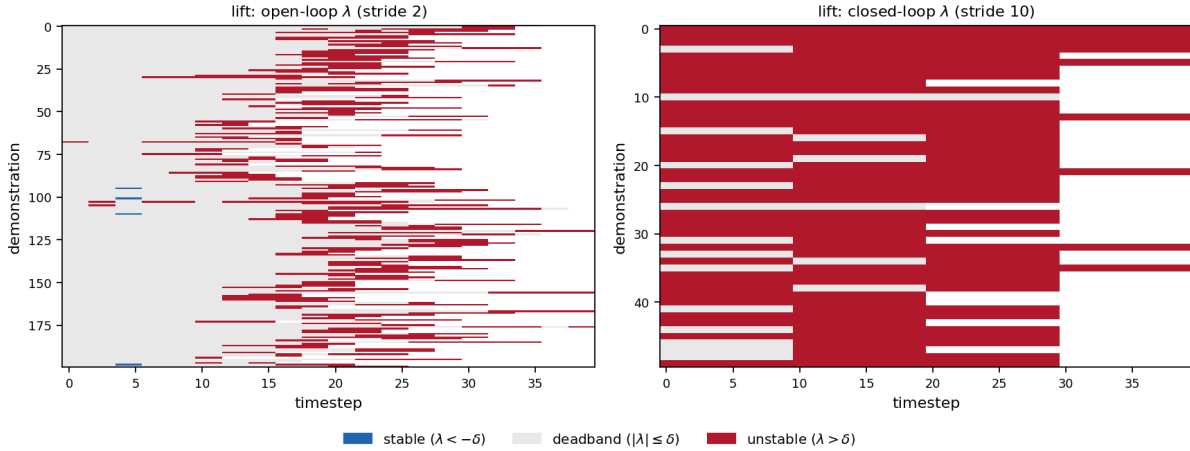
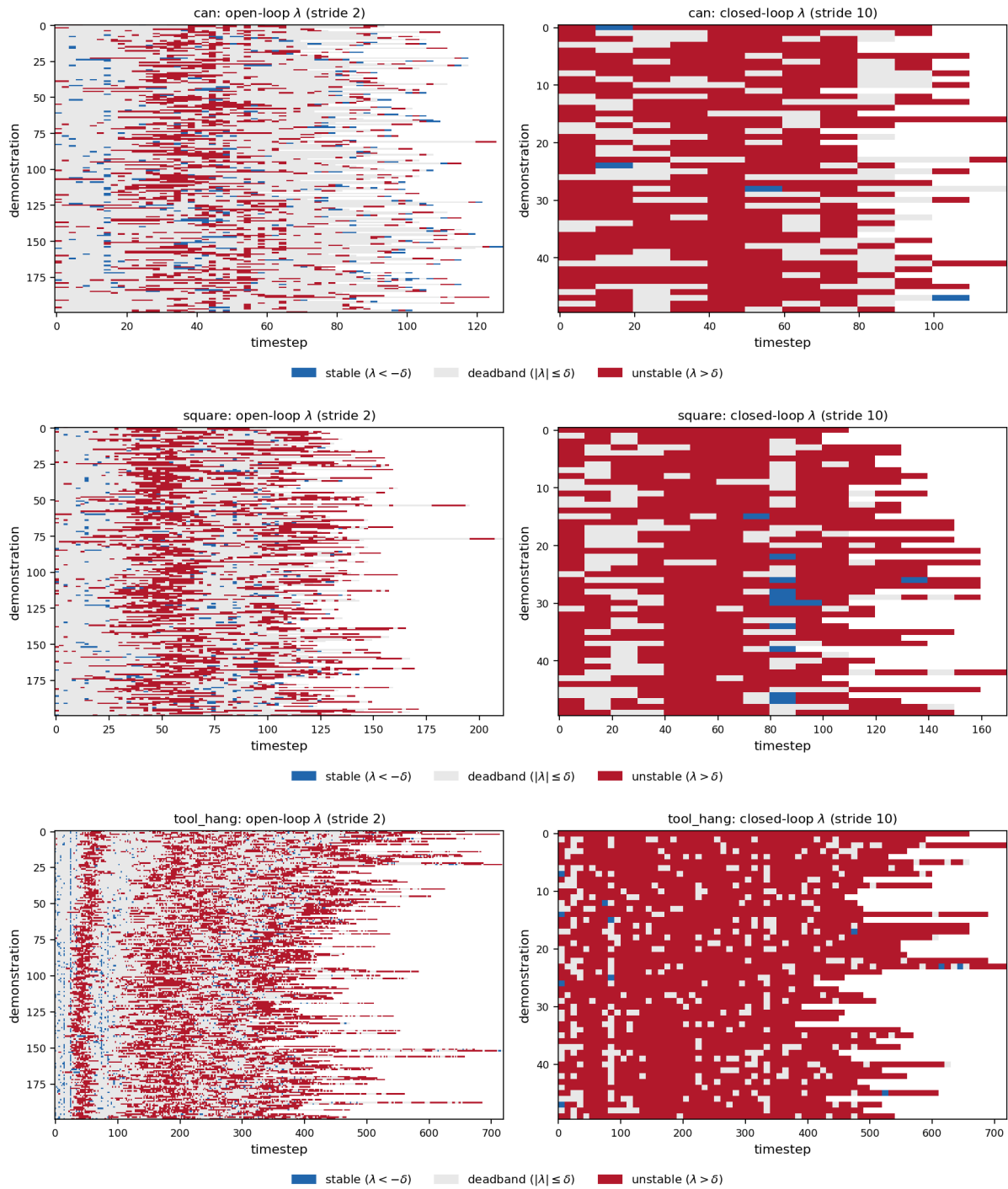


Figure 4: The closed-loop mechanism check on lift. Both histograms measure the same states with the same injected error size. Open loop (blue): replaying the fixed recorded actions leaves λ concentrated at zero; the plant mostly carries the error without amplifying it. Closed loop (magenta): letting the policy replan every step shifts the whole distribution positive (99.3% of stamps), with a second mode near +0.2. The policy’s own corrections are the source of divergence in this mostly-benign task, exactly the feedback-injects-error mechanism the theory predicts.

7 Per-demonstration label timelines: are the labels long-horizon?

A predictor-driven switch is only useful if the labels form segments a controller could act on. If the classes alternated stamp to stamp (salt and pepper), no switching policy could follow them and the whole premise would be in doubt. This section makes that checkable by eye: for every labeled demonstration of the four robomimic tasks, one row per demonstration, timestep on the horizontal axis, and the three-class label at every stamp drawn as a colored block spanning its subsample stride (open-loop stride 2, closed-loop stride 10). Blue is stable ($\lambda < -\delta$), light gray is the deadband ($|\lambda| \leq \delta$), red is unstable ($\lambda > \delta$), with $\delta = 0.0355$ for both channels, and white is past the end of the demonstration.





Contiguity, quantified (a run is a maximal stretch of consecutive same-class stamps within one demonstration; lengths are converted to environment steps):

Task	Channel	occupancy s/d/u	mean run	median run	transitions/demo	mass in runs ≥ 3 stamps
lift	OL	.00/.75/.25	11	8	1.2	0.94
can	OL	.04/.75/.20	9	4	9.8	0.82
square	OL	.03/.63/.34	10	4	11.9	0.86
tool_hang	OL	.02/.58/.40	9	4	52.0	0.83
lift	CL	.00/.14/.86	22	20	0.4	0.64
can	CL	.01/.30/.69	23	20	3.2	0.60
square	CL	.02/.23/.75	27	20	3.8	0.69
tool_hang	CL	.00/.14/.86	43	20	11.6	0.82

What the timelines show, honestly stated:

1. **The labels have real long-horizon structure at the phase level.** Every task shows the same macro-pattern, aligned across demonstrations: an early deadband band during the approach (clearly visible as the gray column on square before step 30 and on tool_hang before step 100), then unstable segments through the contact-rich manipulation phases. This is task structure, not noise, and it is what a switch needs to exist.
2. **They are not salt and pepper, but boundaries flicker.** 82 to 94 percent of open-loop stamp mass sits in runs of three or more stamps (six or more steps). The median open-loop run is still short (4 steps on can, square, and tool_hang), and tool_hang averages 52 class transitions per demonstration, so within the contact phase the signal alternates at the few-stamp scale. This is exactly the behavior the deadband, the median filter, and the rollout-time hysteresis were designed to absorb, and it is why raw stamps should never be consumed without one of those mechanisms.
3. **True stable stamps are rare; the practical contrast is deadband versus unstable.** Only 0 to 4 percent of open-loop stamps fall below $-\delta$. The switch’s “stable” regime in practice is the deadband, which is consistent with how the threshold was chosen (δ at the 95th percentile of the below-median magnitudes).
4. **The closed-loop channel is near-uniformly expansive, in blocks.** The CL panels are mostly red with isolated deadband stamps, matching the 90 to 99 percent positive rates reported above; with only about 54 stamps per demonstration at stride 10, its timelines carry coarser information than the open-loop ones.

One connection to the rollout experiments is worth recording here. The timelines bound how much an oracle or predictor switch can matter: on tool_hang the oracle-switched rollout cells spent only about 16 percent of their steps in the stable mode (44 percent on square, derived from the recorded mean executed k), yet every single rollout episode mixed both modes. The switching signal fires; what limits its value on these tasks is the lopsided occupancy, not a failure to switch.

8 Sanity-check assets: what exists and how to read it

All artifacts live under `results/label_videos/` in the project root (`long-horizon/`), one folder per dataset.

8.1 Annotated demo videos

Files: `results/label_videos/<dataset>/demo_<i>.mp4`, five demos per dataset, for: lift, can, square, tool.hang, mg_stack_d0, mg_square_d0, and each finished LIBERO file (prefix `libero10_`). Each video replays the demonstration in the simulator with a colored border per frame:

- **Red**: the active label at that frame is unstable ($\lambda > +\delta$).
- **Green**: stable ($\lambda < -\delta$).
- **Gray**: deadband (unproven; resolved later by neighbor inheritance).

Figure 6 shows red and gray example frames from four datasets.

The top bar prints the current λ , the verdict, and the window contact flags `c[g t f]` (gripper / table / fixture). Labels are forward-looking: the color at frame t describes the window $[t, t+24]$. What to check when watching: the border should turn red at grasps, insertions, and placements, and stay gray during transport. A `README.json` in each folder repeats the legend.

8.2 Divergence-curve graphs

Files, three most-contractive and three most-expansive stamps per dataset, selected from distinct demos:

`results/label_videos/<dataset>/divergence_curves/
{stable,unstable}_<rank>_demo<id>-t<t0>.png`

A `selection.json` beside them lists each stamp with its label and recomputed λ . Each graph shows the probe at one stamp:

- Thin gray lines: $\|x^{pert}(\tau) - x^{nom}(\tau)\|$ for each of the 8 perturbed branches, on a log scale, over the 24-step window.
- Bold blue line: the branch mean. Dashed orange line: the fitted exponential whose slope is λ . Dotted horizontal line: the injected size σ_u .

Reading: in an expansive stamp the fan of gray lines climbs (the error funnels out); in a contractive stamp it falls back toward zero (the error funnels in); in a neutral stamp it stays flat near the σ_u line. These graphs are the direct visual form of the quantity the label thresholds. Figure 5 shows four representative examples.

9 Caveats and open items

1. **LIBERO labels are provisional.** Interim σ_u (borrowed robomimic mean) and generic contact patterns; contact-alignment AUROC 0.42 to 0.63 where computable, and not computable at all in the six scenes where the generic patterns register contact in every window (those scenes also use the suite-fallback $\delta = 0.019$). Fix: per-scene task-object patterns for the sanity channel, and a measured σ_u once a LIBERO policy is trained. All ten files are labeled.
2. **MimicGen σ_u is borrowed** (square-ph value and robomimic mean). Defensible for same-embodiment same-controller data, but should be re-measured when policies are trained on MimicGen tasks.

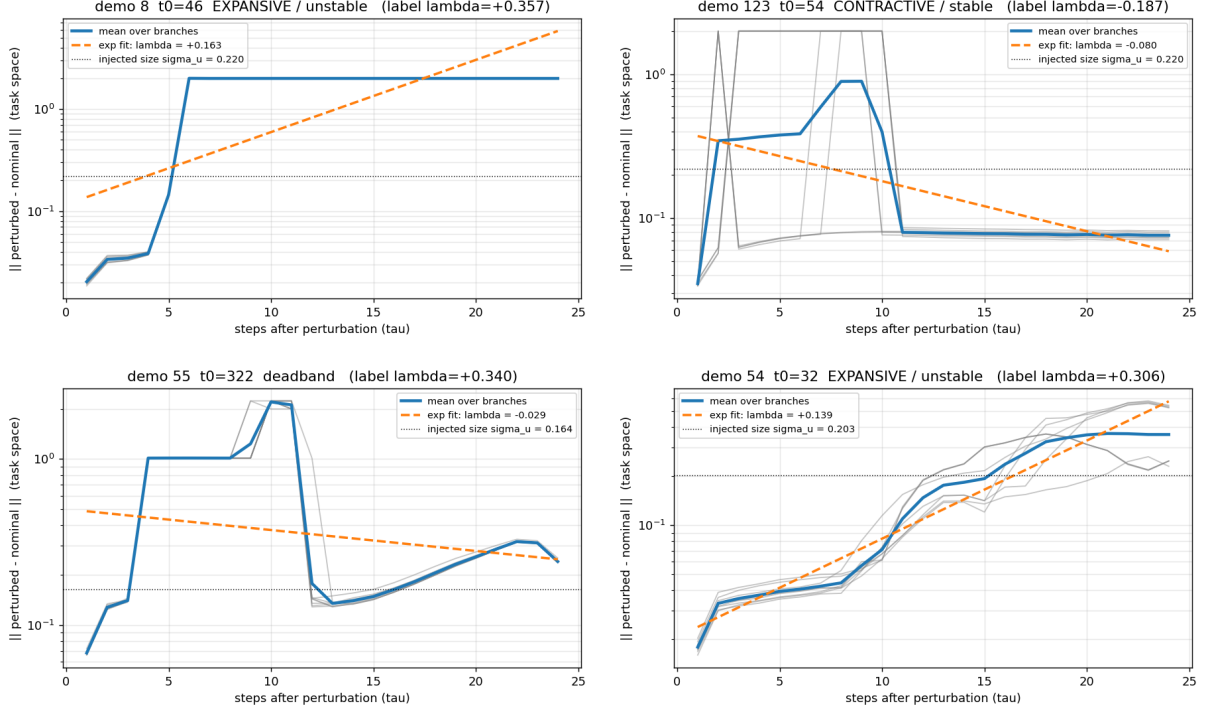


Figure 5: Divergence curves at four labeled stamps: perturbed-versus-nominal error over the 24-step window, log scale. Thin gray lines are the 8 individual perturbed branches, blue is their mean, dashed orange is the fitted exponential whose slope is λ , and the dotted horizontal line marks the injected size σ_u . Top left: square at its most expansive stamp; the error fans out by more than an order of magnitude, the funnel opens. Top right: square at its most contractive stamp; the branches fall back toward the nominal trajectory, the funnel closes. Bottom left: tool_hang at an expansive contact stamp. Bottom right: MimicGen stack_d0 at its most expansive stamp (cube placement); generated demos produce the same signature as human ones.



3. **Closed-loop coverage stops at the four robomimic tasks.** All four sets are done (tool_hang took 53.6 hours of wall-clock). MimicGen and LIBERO have no closed-loop channel yet; adding one costs about a policy-inference-day per task and is not currently scheduled.
4. **can has no free-space stamps** by construction, so its AUROC is undefined and its δ is a fallback. Optional refinement: count object-resting-with-gripper-far as quasi-free for the noise floor.
5. **Contact flags are sanity only.** They never enter the labels. All alignment numbers here are checks of plausibility, not training targets.
6. **FurnitureSim is blocked** on a manual Isaac Gym download.
7. **Deadband dominates** (61–86%). This is expected physics (see Pattern 2) and is handled by neighbor inheritance, but it means the predictor’s decision boundary rests on roughly a quarter to a third of stamps being confidently labeled. Class weighting in the BCE loss is already planned for this.

10 Summary

Sixteen datasets and about 155,000 open-loop stamps are labeled with a verified-deterministic perturb-and-replay FTLE probe, using measured policy error sizes wherever a trained policy exists, plus three finished closed-loop sets. The labels behave the way the hypothesis needs them to: contact-heavy moments are systematically the most expansive, the mostly-stable control task is measurably the most benign, generated demos mirror human demos, and the closed-loop channel shows policy feedback amplifying error exactly where the plant alone would not, now on three tasks (99.3%, 90.4%, and 93.0% of stamps positive on lift, can, and square). The visual sanity layer (annotated videos and divergence graphs) exists for every finished dataset, including all ten LIBERO files. The open-loop labels have since fed the first predictor trainings and the Stage 1a evaluation, which are documented separately.